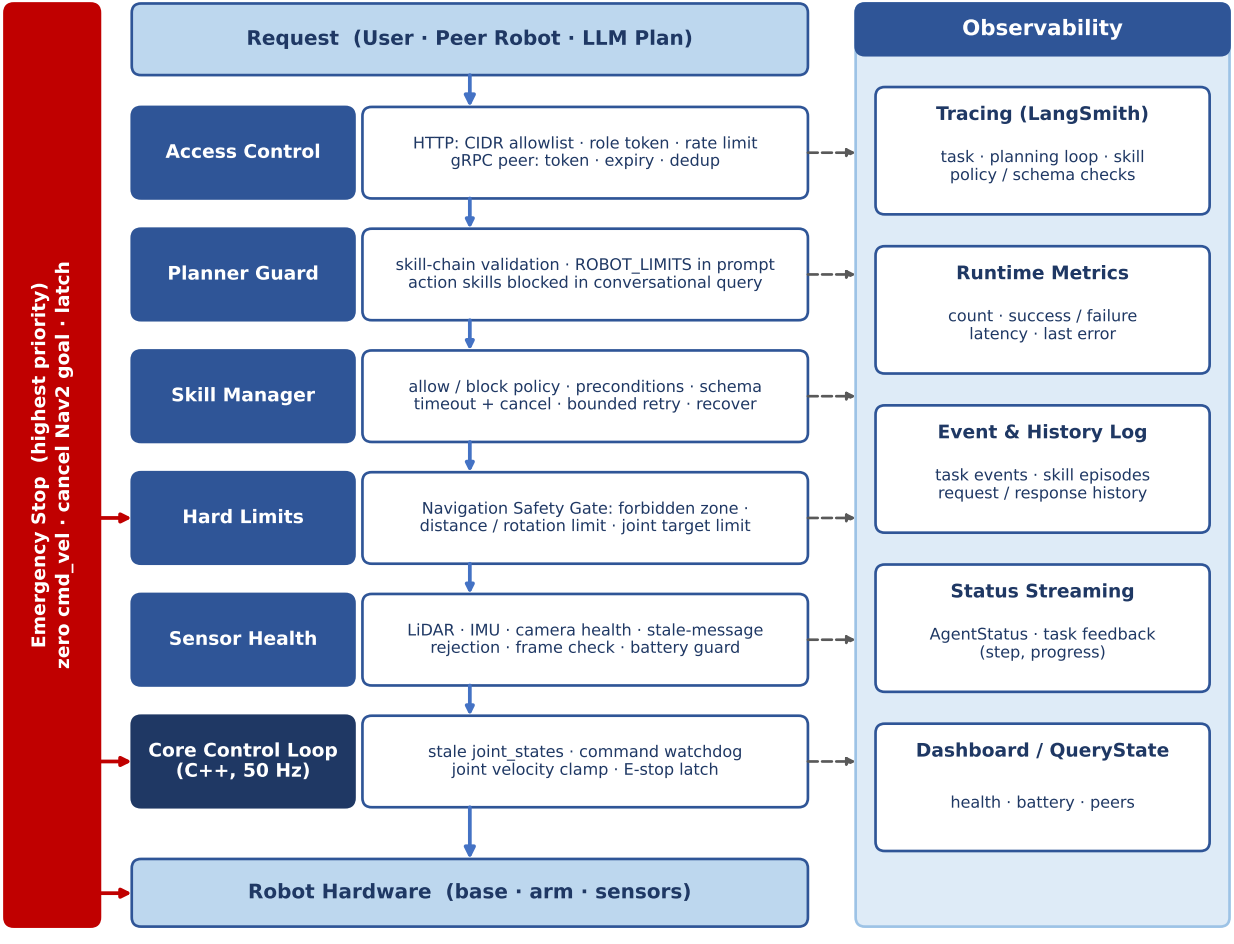
Request (User · Peer Robot · LLM Plan)
Access Control
HTTP: CIDR allowlist · role token · rate limit
gRPC peer: token · expiry · dedup
Planner Guard
skill-chain validation · ROBOT_LIMITS in prompt
action skills blocked in conversational query
Skill Manager
allow / block policy · preconditions · schema
timeout + cancel · bounded retry · recover
Hard Limits
Navigation Safety Gate: forbidden zone ·
distance / rotation limit · joint target limit
Sensor Health
LiDAR · IMU · camera health · stale-message
rejection · frame check · battery guard
Core Control Loop
(C++, 50 Hz)
stale joint_states · command watchdog
joint velocity clamp · E-stop latch
Robot Hardware (base · arm · sensors)
Emergency Stop (highest priority)
zero cmd_vel · cancel Nav2 goal · latch
Observability
Tracing (LangSmith)
task · planning loop · skill
policy / schema checks
Runtime Metrics
count · success / failure
latency · last error
Event & History Log
task events · skill episodes
request / response history
Status Streaming
AgentStatus · task feedback
(step, progress)
Dashboard / QueryState
health · battery · peers